EE1101: Circuits and Network Analysis - Quiz 3

Name: ____________________

Roll No: ____________________

<u>Instructions</u>

1. The exam duration is 35 minutes, starting at 15:40. Ensure that you return the paper by 16:15. Please be aware that **late submissions** may incur **penalties**.
2. Go through all the questions carefully. Any misinterpretations will **not** be considered. Write your solutions in the designated space provided for each question. Answers written outside these areas will not be accepted.
3. Adopting unfair practices will lead to an F grade and will be reported to the disciplinary committee.

I hereby declare that I have read all the instructions carefully and will adhere to them.

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1. (8 points) Given a voltage $v(t)$ (as shown in Fig. 1) applied to a 1 H inductor:

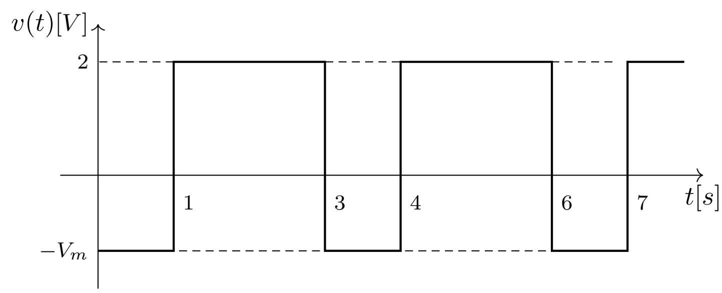

Fig. 1: Voltage applied across an inductor (Question 1)

   (a) (2 points) Find the value of V_m that results in the inductor current having a steady average value over time. ________.

   (b) (2 points) For the V_m value determined in part (a), find the average value of the inductor current. ________.

   (c) (2 points) What is the maximum value of inductor current? ________.

   (d) (2 points) What is the average value of instantaneous power associated with the inductor? ________.

   Note: Please write "NO", if you feel that a steady average current cannot be achieved.

2. (2 points) Consider the circuit shown in Fig. 2. Let Q_1 denote the reactive power of the inductor and Q_2 denote the reactive power of the capacitor. The value of $Q_1 + Q_2$ is ________.

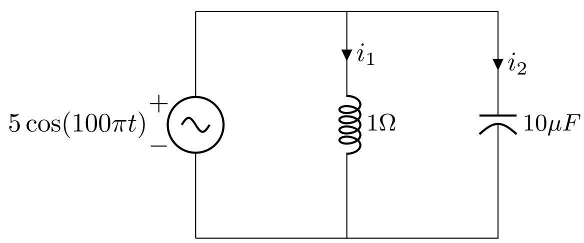

Fig. 2: Circuit for question 2

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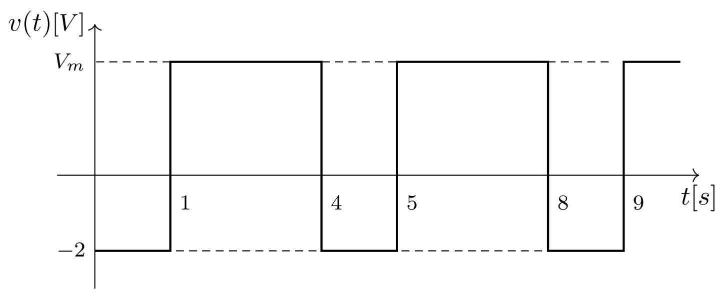

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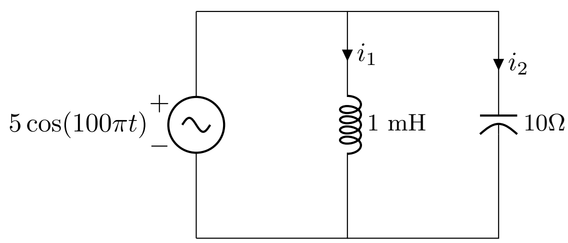

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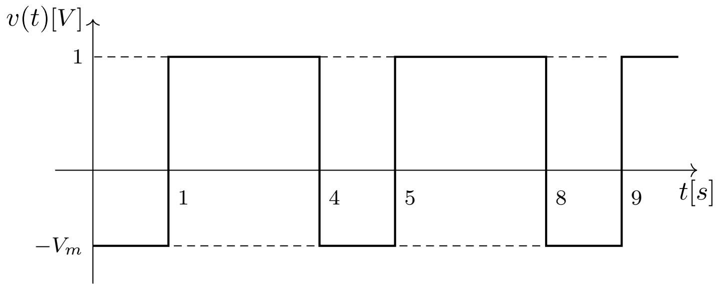

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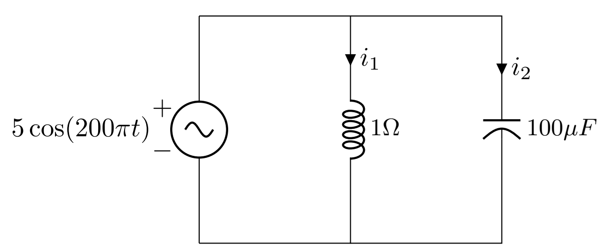

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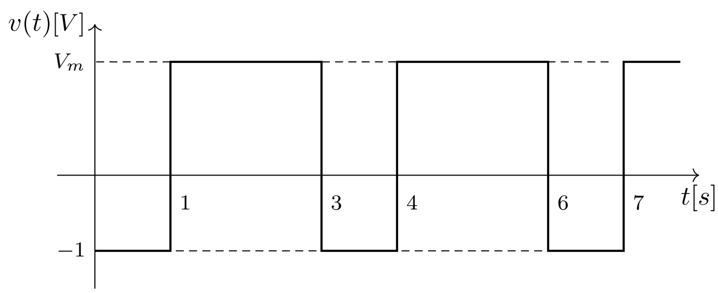

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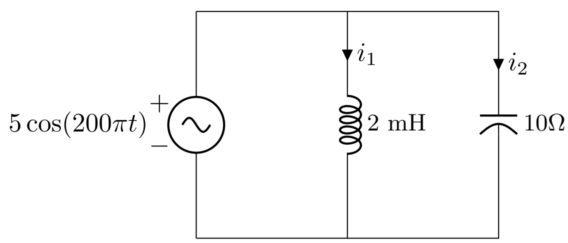

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